

Comprehensive Briefing on Cohesive Sediment Transport Mechanics

Executive Summary

Cohesive sediment transport is a multi-physics regime governed by the interplay of hydraulic forces, electrochemical interactions, and biological processes. Unlike noncohesive sediments (such as sand), which are primarily influenced by gravity and fluid shear, cohesive sediments—typically comprising silts and clays with diameters less than $63\ \mu\text{m}$ —exhibit behaviors dominated by inter-particle forces.

Key Takeaways

- **Flocculation Dynamics:**
Fine particles aggregate into flocs, which are porous, fractal structures. Their size and settling velocity are controlled by a balance between aggregation (particle collisions) and breakup (turbulence).
- **Non-Linear Settling:**
Settling velocity is not a simple function of size; it is influenced by sediment concentration, salinity, and turbulence, often exhibiting a two-stage "inhibition–acceleration" process.
- **Probabilistic Deposition and History-Dependent Erosion:**
Deposition is modeled as a probability based on bed shear stress, while erosion thresholds are not unique but depend on the history of consolidation and bed aging.
- **The 10–20% Threshold:**
Mixed sediments begin to exhibit cohesive behavior when the clay/mud fraction exceeds approximately 10–20%, leading to the suppression of bedforms like ripples.

1. Distinctive Features of Cohesive Sediments

Cohesive sediments represent a fundamentally different transport regime from sand-bed transport. Their behavior is governed by electrochemical forces (van der Waals attraction and electrostatic forces) and organic bonding rather than just gravity.

1.1 Composition and Cohesion

- **Classification:**

Generally defined as sediments where:

$$d < 63 \mu m$$

(silt and clay)

- **Threshold for Cohesion:**

Strong cohesion typically emerges when the clay fraction exceeds approximately 10%.

- **Mohr–Coulomb Concept:**

The resistance of these sediments is represented by the cohesion law:

$$\tau = c + \sigma \tan \phi$$

where:

- c : cohesion intercept (often analogous to the yield stress, τ_y , measured in rheological tests on the same sediment)
- ϕ is the internal friction angle

This allows the sediment to resist motion even when effective stress (σ) approaches zero.

1.2 Comparison of Sediment Types

Feature	Noncohesive (Sand/Gravel)	Cohesive (Clay/Silt)	Mixed Sediments
Dominant Forces	Gravity vs. fluid shear	Electrochemical, biological, gravity	Transition of both
Particle Interaction	Independent grains	Strong inter-particle bonding	Partial bonding
Transport Mode	Bed load + suspended load	Flocs (aggregates)	Hybrid behavior
Threshold (Motion)	Well-defined (Shields curve)	Ill-defined, history-dependent	Transitional
Deposition	Deterministic	Probabilistic (Krone model)	Depends on clay fraction
Time Dependence	Minimal	Strong (aging, consolidation)	Moderate

2. Flocculation Mechanics

Flocculation is the process by which clay particles form larger aggregates called flocs. This process is dynamic, with flocs constantly undergoing aggregation and breakup.

2.1 Environmental Influences

- **Salinity:**
In freshwater, particles tend to repel each other due to negative charges. In saline water, cations (such as Na^+ and Ca^{2+}) reduce this repulsion, significantly increasing flocculation.
 - **Turbulence:**
Moderate turbulence promotes collisions and aggregation; however, high turbulence leads to the mechanical breakup of flocs.
 - **Floc Properties:**
 - Microflocs: $< 133 \mu\text{m}$
 - Macroflocs: $> 133 \mu\text{m}$
 - Density is much lower than mineral density
 - Density decreases as floc size increases
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2.2 Relationship of Density and Size

Floc density is empirically related to size through the expression:

$$\rho_{floc} - \rho = a_f D_{floc}^{-b_f}$$

Coefficients a_f and b_f vary by location (e.g., $b_f = 0.97$ in Chesapeake Bay vs. 1.09 in San Francisco Bay), reflecting the site-specific nature of cohesive sediment behavior.

3. Vertical Structure and Settling Velocity

3.1 Layered Suspension Structure

Under weak flow conditions, cohesive sediment suspensions form a distinct vertical structure:

1. Mixed Suspension Layer: Well-mixed top layer
 2. Stratified Suspension Layer: Increasing concentration with depth
 3. Lutocline: A sharp gradient or shear layer
 4. Fluid Mud: Behaves as a non-Newtonian fluid with no effective stress
 5. Consolidated Bed: Measurable effective stress where particles are in permanent contact
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3.2 Settling Velocity Controls

The floc settling velocity (w_{sf}) is a product of several correction factors applied to the primary particle settling velocity (w_s):

$$w_{sf} = K_d K_s K_t K_{sa} w_s$$

where:

- K_d : size effect
 - K_s : concentration effect
 - K_t : turbulence effect
 - K_{sa} : salinity effect
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The Role of Concentration and Salinity

Settling is not monotonic and follows three concentration regimes:

1. Free Settling: Low concentration
 2. Flocculation-Enhanced Settling: Moderate concentration increases collision and settling speed
 3. Hindered Settling: High concentrations slow the settling process
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Two-Stage Salinity Effect (Li et al., 2025)

- **Stage 1 (Inhibition):** At very low salinities (0–2 ppt), initial increases in fluid density and viscosity slightly slow settling. This effect is relevant only in freshwater and brackish systems.
- **Stage 2 (Acceleration):** At higher salinities, EDL (electrical double layer) compression enhances flocculation and drastically increases settling rates. In typical estuarine and coastal environments (5–35 ppt), this mechanism dominates, and the net effect of salinity is strongly acceleratory, with Stage 1 inhibition being negligible.

4. Deposition and Erosion Processes

4.1 Deposition (Krone Model)

Deposition is a probabilistic process rather than a binary one. The deposition rate is:

$$D_b = \alpha_b w_{sf} C$$

where α_b depends on bed shear stress (τ_b):

- Full Deposition: $\tau_b < \tau_{bd,min}$
- No Deposition: $\tau_b > \tau_{bd,max}$
- Partial Deposition:

$$\alpha_b = 1 - \frac{\tau_b - \tau_{bd,min}}{\tau_{bd,max} - \tau_{bd,min}}$$

4.2 Erosion (Partheniades Model)

Erosion occurs when bed shear stress exceeds the critical shear stress (τ_{ce}), which depends on density, consolidation state, and clay content.

- Surface Erosion: Individual flocs are removed
 - Mass Erosion: Occurs when τ_b is significantly higher (typically 2–5 times τ_{ce})
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Erosion Rate Equation

$$E_b = M \left(\frac{\tau_b}{\tau_{ce}} - 1 \right)$$

where M is the erosion properties constant.

5. Consolidation and Mixed Sediments

5.1 Consolidation

Consolidation is the time-dependent compaction of deposited mud. As pore water is expelled:

- density increases
- strength increases

making the bed more resistant to future erosion.

5.2 Mixed Sediment Behavior

The transition from noncohesive to cohesive behavior is critical for natural river modeling.

- **Ripple Suppression:**
As the mud fraction approaches 10–20%, bedforms such as ripples are suppressed and reduced in height.
 - **Increased Resistance:**
Higher clay content increases the critical shear stress required for transport and reduces the overall transport rate.
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5.3 Biological Effects

Biological factors, specifically biofilms and Extracellular Polymeric Substances (EPS), act as a "sticky substance" that:

- increases particle cohesion and forms bio-flocs
- reduces erosion and transport rates
- suppresses bedform development